

Digital Logic Design (EE1005)

Course Instructor(s):

Ms. Aneeqa Khalil

Section: BSCY A&B

Final Exam

Total Time (Hrs): 3

Total Marks: 110

Total Questions: 5

Date: Dec 15, 2025

241-2096

Roll No

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CV-A

Course Section


Student Signature

Attempt all the questions.
Attempt all the questions on Answer Sheet.

[CLO 3: Analyze small-scale combinational and sequential digital circuits]

Q1. Draw and label the circuit diagram of a universal shift register using four 4 x 1 multiplexers and four D flip flops. The shift register should operate according to the following table. [10 marks]

✓

S1	S0	Operation
0	0	Shift Left
0	1	2's complement inputs
1	0	Hold Value
1	1	Incremental

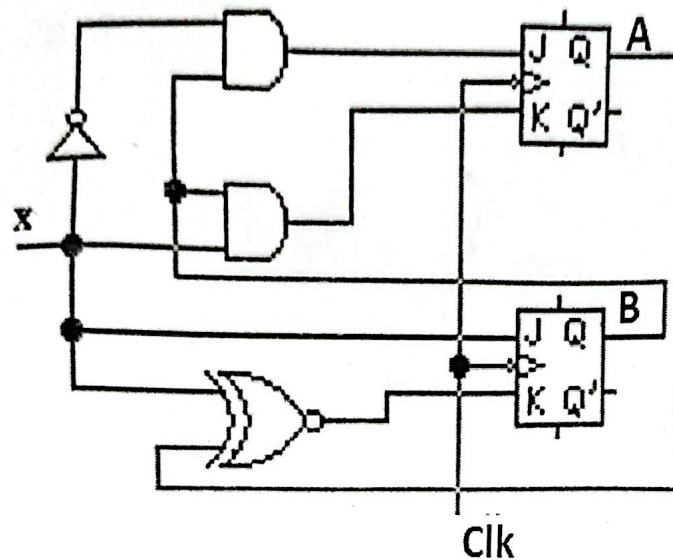
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[CLO 3: Analyze small-scale combinational and sequential digital circuits]

Q2. Analyze the following circuit with JK flip flops and provide the flip flop input equations, state table and state diagram. [10 marks]

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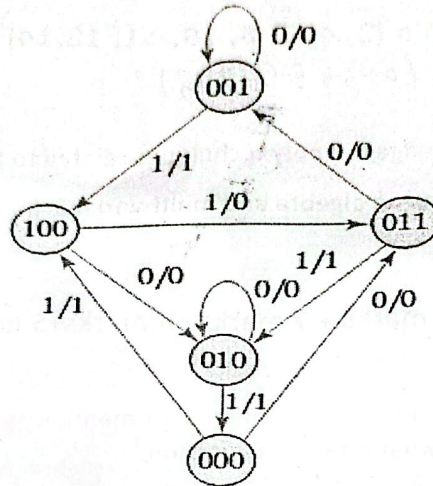
[CLO 2: Demonstrate the acquired knowledge to apply techniques related to the design and analysis of digital electronic circuits including Boolean algebra and multi-variable Karnaugh map methods]

Q3.

[10 marks + 12 marks + 10 marks]

1. Students of Batch BSCS-2024 have developed a new flip-flop named AB Flip-Flop. When inputs of A and B are 00, 01, 10 and 11, the flip-flop is complemented, resets to 0, set to 1 and no change occurs respectively. For AB Flip-Flop:
 - i. Tabulate its characteristic table
 - ii. Tabulate excitation table
 - iii. Derive its excitation table
 - iv. Construct AB-type Flip-Flop using JK flip-flop

2. Design a sequential circuit for following State transition diagram using T flip-flops. This circuit has three flip flops A, B, C; one input x and one output y. Treat unused states as don't care.



3. Draw and label circuit of serial adder.

[CLO 1: Identify and explain fundamental concepts of digital logic design including basic and universal gates, number systems, binary coded systems, basic components of combinational and sequential circuits.]

Q4.

[8 marks + 12 marks +8 marks]

1. Draw 4-bit adder- subtractor circuit and label following values of M, A and B and find Sum and carry

Outputs: $M = 1$, $A = 1011$, $B = 1101$

$A - B$

2. Design a 4-bit priority encoders with following priority sequence.

$D1 > D2 > D0 > D3$

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3. Implement following function 4 x 1 Multiplexer. Construct Multiplexer table and use selections lines appropriately.

$$F(A, B, C, D) = \pi(2, 4, 7, 8, 10, 11, 13, 14) \quad 8$$

$$\Sigma(0, 1, 3, 5, 6, 9, 12, 15)$$

[CLO 2: Demonstrate the acquired knowledge to apply techniques related to the design and analysis of digital electronic circuits including Boolean algebra and multi-variable Karnaugh map methods]

Q5.

[10 marks + 7 marks + 5 marks + 5 marks + 3 marks]

1. Simplify the following Boolean function F by Kmap, and mention the prime implicants and essential prime implicants, and what is the final function.

$$F(A, B, C, D) = \prod(4, 12, 7, 2, 10) \quad 10$$

$$d(A, B, C, D) = \prod(0, 6, 8)$$

2. Convert the following expression into sum of products and product of sums

$$(u + xw)(x + uv) \quad 6$$

3. Draw the logic diagram corresponding to the following Boolean expression without simplifying it

$$(AB + A'B')(CD' + C'D) \quad 5$$

4. Simplify the following Boolean expressions to a minimum number of literals by algebraic rules:

$$(BC + AD)(AB + CD) \quad 1$$

5. What is the largest binary number that can be expressed with 16 bits? What are the equivalent decimal and hexadecimal numbers? 3